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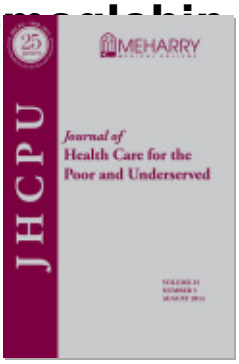


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Journal of Health Care for the Poor and Underserved, Volume 24, Number 4, November 2013, Supplement, pp. 48-60 (Article)

Published by The Johns Hopkins University Press
*DOI: 10.1353/hpu.2014.0003*

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# A Community-Based Pilot Study of a Diabetes Pharmacist Intervention in Latinos: Impact on Weight and Hemoglobin A1c

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*Abstract:* Type 2 diabetes disproportionately affects Latinos increasing their risk of diabetes-related complications. This study used a randomized controlled design with a community-based approach to evaluate the impact of a culturally tailored pharmacist intervention on clinical outcomes in Latino diabetics. The intervention included a focused discussion and two individual pharmacist counseling sessions on medication, nutrition, exercise, and self-care to promote behavior changes. Sessions were culturally adapted for language, diet, family participation, and cultural beliefs. Clinical outcomes were measured at baseline and three months. Nineteen intervention and 24 control participants completed the study. Mean BMI reduction was greater for intervention than for control group participants (–0.73± 0.07 kg/m$^2$ versus + 0.37±0.02 kg/ m$^2$ p<.009 respectively). Hemoglobin A1c was significantly reduced by 0.93±0.45% in the intervention group only. There was no significant difference in blood glucose, blood pressure, or lipid levels. An innovative culturally-sensitive pharmacist intervention improved selected clinical outcomes among Latino diabetics.

*Key words:* Pharmacist, type 2 diabetes, Latinos, intervention, weight loss, Hemoglobin A1c.

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There is mounting evidence that type 2 diabetes (T2DM) disproportionately burdens the elderly and minority groups in the United States.[1,2] Latino diabetics have higher blood sugar levels and higher incidence of undiagnosed diabetes than non-Hispanic Whites,[3] and are more likely to have hemoglobin A1c (HbA1c) levels of more than 11%.[4] Such ethnic disparities experienced by Latinos increases their risk of diabetes-related complications like retinopathy, neuropathy, kidney disease, and amputations. Florida is the state with the third highest concentration of Latinos; Florida's Latino population makes up a much higher proportion of the state's entire population than Latinos make up of the U.S. population as a whole (23% *vs.* 16.3%, respectively).[5,6] Furthermore, South Florida Latinos differ from Latinos in other states in terms of country of origin, with a higher prevalence of Caribbean and Central and South American countries of origin. The Latino population in Florida is also 49% foreign-born compared with 37.1% in the U.S. population.[7] This profile of a large and diverse Latino population with a high incidence of diabetes creates a burden requiring culturally sensitive diabetes care to avoid health complications and associated increases in health care costs. Hundreds of interventions on diabetic patients have been reported in the literature in the past decade. Among those, pharmacist interventions delivering medication therapy management and diabetes education have been successful at significantly reducing HbA1C.[8–10] The results of these studies vary depending on the setting, the capacity of the pharmacist to modify/prescribe drug therapies, and intervention length. However, there is a gap in the literature regarding culturally sensitive interventions by pharmacists when delivering care to Latino patients.[11] For example, when treating Latinos, the health care provider must be aware of language barriers, health literacy, quality-of-care factors, self-care behaviors, biological differences, and socioeconomic status issues including access to the health care system. The development of linguistically and culturally appropriate interventions that promote diabetes prevention and improve clinical outcomes in current diabetic patients should be a priority.

In addition to the individual's beliefs and behaviors, self-care practices in patients with T2DM are affected by three major social and environmental factors: 1) practitioner/health care system, 2) community/work environment, and 3) the patient's family.[11] The impact of the patient's family in the management of T2DM patients is the least studied.[12] It has been observed that as family support in healthy eating and exercise behavior increases, perceived barriers to self-care decrease.[13,14] These findings suggest that diabetes interventions should target both the participant and family to encourage appropriate nutrition, exercise, self-management, medical treatment, involvement, and support.

The objective of this study was to determine the effectiveness of a community-based, culturally sensitive pharmacist intervention at improving weight, body mass index (BMI) and HbA1c among immigrant Latinos residing in Broward County, Florida. This study used a community-academic partnership design recruiting Latino diabetics and their support people seeking services at Hispanic Unity of Florida (HUF), a community-based organization. The diabetes literacy and self-efficacy screening and training project (DLASST) designed a Latino pharmacist intervention to evaluate the impact of diabetes education, nutrition, exercise, self-monitoring of blood glucose, and

medication therapy management on clinical outcomes. In this study, *Latino* refers to people of Cuban, Mexican, Puerto Rican, South or Central American, or other Spanish culture or origin, regardless of race.

## Methods

**Community-academic partnership**. The Hispanic Unity of Florida (HUF) and Nova Southeastern University (NSU) College of Pharmacy (COP) formed a community-academic partnership to address poor outcomes from T2DM among Latinos. The HUF is a community-based organization whose services aim to ease the transition of immigrants and promote empowerment and self-sufficiency to all underserved populations from different ethnic and racial backgrounds. Nova Southeastern University is a private non-for-profit university of over 29,000 students; the COP graduates over 250 pharmacy students per year, most of Hispanic origin. Type 2 diabetes was identified by HUF as a high priority health concern for the Latino community of Broward County. The study took place at HUF sites and at NSU COP's clinic pharmacy and was approved by the NSU Institutional Review Board and the Office of Human Subjects Protection, Washington, D.C.

**Study design**. This pilot study used a randomized control trial design to test the efficacy of the Pharmacist-centered Assessment and Reinforcement of Diabetes Self-efficacy (PARDS) intervention on clinical outcomes (weight, body mass index, HbA1c, random blood glucose, blood pressure and lipid levels).

**Study participants**. A total of 84 participants were recruited from those seeking services at HUF. A total of 43 self-categorized Latino adults with type 2 diabetes completed all phases of the study. Participants were recruited using study flyers that were distributed at HUF programs and posted at their facilities. Recruitment ran between January and July 2010 and the trial was closed on November 2010. The diagnosis was confirmed in a clinical screening using the American Association of Diabetes criteria.[15] Latino diabetics with or without a support person (either a family member or friend) were included. Participants with current gestational diabetes, inability to watch a video or participate in a focused discussion group due to hearing impairments were excluded. Participants received $10 gift cards for each visit to compensate for time and travel. All participants were clinically screened at baseline prior to randomization into the control or intervention groups. If randomized to the intervention group, patients were asked to come back for three educational sessions every two weeks during the first six weeks and a follow-up clinical screening three months later. The control group had the same timeline but no educational sessions were given. The support person was asked to come with the study participant for every clinical screening and educational session, as appropriate.

**Baseline and follow-up screenings**. Most clinical screenings were held at HUF and some at the NSU's clinic pharmacy. After obtaining informed consent, baseline and three-month clinical screenings consisted of a clinical assessment, a pharmacist clinical debriefing, and information about community resources.

As part of the clinical assessment, Latino pharmacists and supervised student pharmacists conducted a standardized physical assessment to both participant and support

person which included recording the weight, height, waist circumference, BMI, and blood pressure. Random blood samples were collected by finger stick and analyzed *in situ* to determine blood glucose, HbA1c, cholesterol levels, and triglycerides. Additionally, a urine spot sample was collected once to quantify albumin/creatinine ratio using a urine stick analyzer. HbA1c and lipids were measured using capillary blood samples on the DCA Vantage Analyzer; SIEMENS-HBA1c; UK and Cholestech LDX Analyzer LIPID ANALYZER; USA/CA. After participating in a debriefing session and receiving explanation about their clinical values, all participants were given two copies of their clinical results; one for their primary health care provider and one for their records.

**Pharmacist Assessment and Reinforcement of Diabetes Self-management**. The Pharmacist Assessment and Reinforcement of Diabetes Self-management (PARDS) consisted of an intensive and multi-faceted half-day Saturday morning session and two sixty-minute individual consultations with a pharmacist convened two, four and six weeks post-baseline. The half-day session consisted of four parts: (1) A 90-minute focused discussion group (FDG) on T2DM knowledge, beliefs, and barriers and motivators to clinical and self-management; (2) a 13-minute animated video titled, 'What is Diabetes/¿Qué es la Diabetes?' by Animax Health, 2006 Health LAMP, Inc.; (3) a 45-minute self-monitoring of blood glucose training session, with (4) new monitors and strips distributed to participants.

A FDG is a qualitative method classified as using participant-observer methodology.[16] Like traditional *focus groups,* FDGs are intended to collect qualitative data for research purposes but they are also intended to teach participants and dispel myths about the health topic under discussion. Closing discussion points were led by the pharmacist and focused on misconceptions brought up by participants and their partners during the FDG. This method alone has been shown to be an effective educational intervention among Latinos.[17] All groups were convened in Spanish and headed by the project coordinator (Master in Public Health level), an experienced moderator, and the pharmacist's role was that of a participant observer. Session I was followed by two individual pharmacist counseling sessions (Sessions II and III) conducted two weeks and four weeks after session I.

Individual Sessions II and III used three synergistic methodological approaches: 1) predetermined action items[18,19]; 2) Teach Back Method[20] and 3) action planning.[18,19] Discussion of predetermined action items included the Brown Bag Test of medication knowledge and adherence, frequency of self-monitoring of blood glucose (SMBG), nutrition, and physical activity. Nutrition training that used Latin American food replicas (Fort Atkinson, WI: Nasco, 2004) and the Food Plate method of portion size determination[20,21] were used in these sessions. The goal of each session was to open meaningful discussions for each predetermined action item in the context of the diabetic participant's perceived importance of each, to dispel misconceptions and myths regarding diabetes management, and to reinforce the importance of action items. The Teach Back Method entails asking participants to repeat salient points of discussions about each action item which has been demonstrated to ensure comprehension. Action planning helps to establish and maintain new behaviors by being tailored to the individual diabetic patient's life (taking into account, for example, socioeconomic status). Pharmacists assisted participants in identifying components of each action item that the

participant felt they would achieve through behavioral change by choosing achievable short-term goals. Once a reasonable and agreed upon goal was reached, the pharmacist applied the Teach Back Method for action planning.

**Statistical analysis**. Chi-square and independent t-tests were used to evaluate differences in baseline characteristics and clinical outcomes across groups, where appropriate. A linear regression model was used to evaluate the mean change in clinical outcomes from baseline to three-month follow-up controlling for group allocation, demographic characteristics (gender, age, language spoken at home, religious affiliation, employment status, being retired, insurance status and type, marital status, country of birth, number of years living in the U.S., and income and education level), having a partner at study baseline, and prior one-on-one or group diabetes education, if any were different across groups at baseline. An alpha of 5% was considered significant. Results will be presented as percentages or mean and standard error of the mean, as appropriate.

## Results

**Recruitment and retention**. Recruitment into the study was difficult among Latinos who sought social and other services at HUF. Of 84 people self-identified as Latino and having T2DM, only 43 completed all phases of the study: 24 control and 19 intervention group participants. Time conflicts with home and job, health, and transportation were the most common reasons for not returning to group or individual sessions and follow-up screening.

**Participant characteristics**. The majority of participants were women of Caribbean, Central American, or South American nationalities. Most spoke Spanish language at home and were married or partnered. Approximately half of the subjects were unemployed and/or uninsured (Table 1). Education, income level and length of time living in the U.S. were significantly unbalanced across groups. All regression analyses were controlled for these three covariates. Sixty-seven percent of control group and 79% of intervention group participants presented with a support partner at baseline. However, 33% and 42% of the partners in the intervention and control groups respectively came with the study participant to both clinical screenings. Partners were a spouse, friend, or first-degree relative (e.g., sibling, child). Most partners were female and had an average age of 48 (20 to 77) years.

**Clinical outcomes**. *Weight and body mass index.* Mean control and intervention group weights were 81.4 ± 3.3 kg and 87.4 ± 3.9 kg at baseline, respectively. After three months, weights in control and intervention groups changed to 82.4 ± 2.9 kg and 85.5 ± 3.8 kg, respectively (Table 2). Weight loss was significantly greater among intervention group participants compared with the control group at follow-up (mean difference of −2.7 kg p=.045 (Figure 1). Ninety-one percent of all participants did not meet American Diabetes Association (ADA) target body mass index (BMI) of less than 24.9 kg/m$^2$ at baseline. Twenty eight percent were classified as overweight (BMI=25–29.0 kg), 33% as obese (BMI=30.0–34.0 kg) and 30% were morbidly obese (BMI> 35) with a mean BMI of 32.6 ± 2.6 kg/m$^2$ for all participants at baseline (n=43). Compared with baseline measurement there was a significant reduction in BMI at three months in the intervention group *versus* the control (mean difference across groups 1.1 kg /m$^2$, p=.009).

# Table 1.

## PARTICIPANT SOCIO-DEMOGRAPHIC CHARACTERISTICS

| | Control | Intervention (PARDS) |
|---|---|---|
| | Mean ± SD or % N=24 | Mean ± SD or % N=19 |
| Age, yrs | 55±10 | 54±9 |
| Age category | | |
| <65 | 75% | 89% |
| ≥65 | 25% | 11% |
| Gender | | |
| Female | 79% | 58% |
| Male | 21% | 42% |
| Marital Status | | |
| Single | 17% | 5% |
| Married or partnered | 58% | 68% |
| Separated/divorced/widowed | 25% | 26% |
| Income category[a]($) | | |
| <20,000 | 86% | 31% |
| ≥20,000 | 14% | 69% |
| Education level[a] | | |
| No or some school/HS diploma/GED | 67% | 16% |
| Some college/college degree | 33% | 84% |
| Employment Status | | |
| Employed full time | 28% | 29% |
| Employed part time | 17% | 29% |
| Unemployed | 56% | 41% |
| Retired or disabled | | |
| No | 88% | 89% |
| Yes | 13% | 11% |
| Insurance Status | | |
| Uninsured | 43% | 47% |
| Private insurance | 9% | 11% |
| Public insurance | 22% | 5% |
| Other community or discount health program | 26% | 37% |
| Region of birth | | |
| Caribbean | 33% | 11% |
| Central America | 13% | 11% |
| South America | 42% | 74% |
| Other | 13% | 5% |
| Living in the U.S.[a] | | |
| 10 years or less | 33% | 72% |
| Greater than 10 years | 67% | 28% |
| Language spoken at home | | |
| Spanish Only | 71% | 89% |
| English Only/ Both English and Spanish | 29% | 11% |

[a]Education, income level and length of time living in the U.S. were significantly different across groups, $p<.05$

PARDS=Pharmacist-centered Assessment and Reinforcement of Diabetes Self-efficacy

## Table 2.

### DEMOGRAPHIC CHARACTERISTICS IN CONTROL AND INTERVENTION GROUPS

| | Control | | Intervention | |
|---|---|---|---|---|
| | **Baseline** | **Follow-up** | **Baseline** | **Follow-up** |
| Weight (kg) | 81.4 ± 3.3 | 82.4 ± 2.9 | 87.4 ± 3.9 | 85.5 ± 3.8[a] |
| BMI | 31.5 ± 1.3 | 31.9 ± 0.9 | 31.2 ± 1.9 | 30.5 ± 1.4[a] |
| Waist Circumference (cm) | 105.1 ± 2.5 | 104.7 ± 2.2 | 111.1 ± 4.01 | 103.8 ± 3.68 |
| Hemoglobin A1C (%) | 8.2 ± 0.4 | 8.0 ± 0.2 | 8.3 ± 0.4 | 7.3 ± 0.3[b] |
| Random Blood Glucose (mg/dL) | 183 ± 15 | 202 ± 20 | 190 ± 23 | 165 ± 16 |
| Systolic BP (mmHg) | 131 ± 4.0 | 126 ± 3.4 | 129 ± 3.2 | 126 ± 2.8 |
| Diastolic BP (mmHg) | 81.8 ± 2.4 | 80.1 ± 2. | 82.2 ± 2.0 | 79.8 ± 2.0 |
| Total Cholesterol (mg/dL) | 180 ± 8.8 | 175 ± 8.6 | 179 ± 7.1 | 169 ± 8.9 |
| LDL Cholesterol (mg/dL) | 88.2 ± 5.5 | 96.8 ± 5.4 | 99.9 ± 14 | 91.0 ± 8.8 |
| HDL Cholesterol (mg/dL) | 37.1 ± 2.62 | 38.3 ± 2.6 | 34.8 ± 3.7 | 37.3 ± 2.4 |
| Triglycerides (mg/dL) | 212.7 ± 38 | 228 ± 38 | 211 ± 47 | 198 ± 26 |

[a]Follow-up differences are statistically significant from baseline compared with the control group, p<.05
[b]Follow-up difference is statistically significant from baseline, p<.05

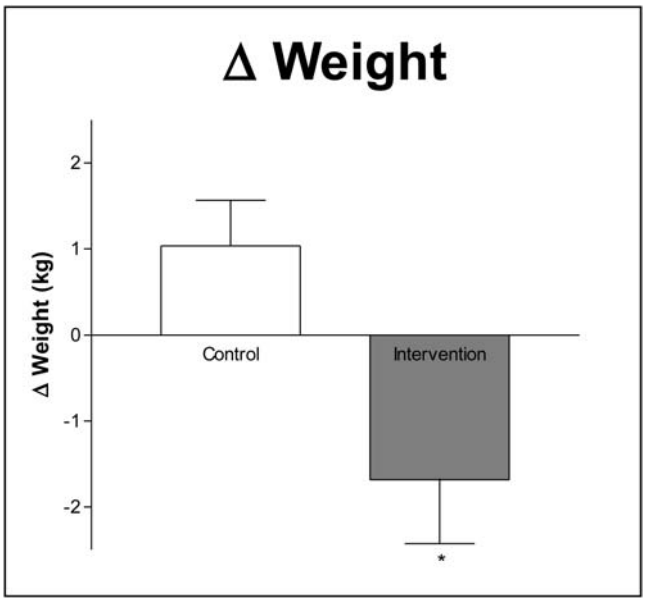


Figure 1. Change in weight (kg) from baseline to follow-up among control and intervention groups (*p= 0.009 from baseline).

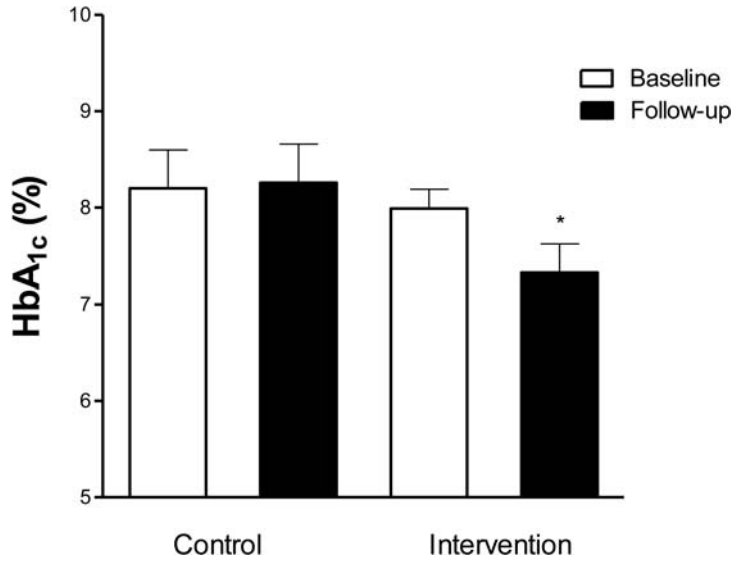


Figure 2. Hemoglobin A1c levels at baseline and at follow-up among control and intervention groups (*= p<.05 from baseline).

*Glycosylated hemoglobin (HbA1c).* There was no significant difference between groups in HbA1c levels at baseline (8.2 ±0.4 versus 8.3 ±0.4). Follow-up mean HbA1c levels were 8.0 ± 0.2 and 7.33 ± 0.3 for control and intervention group participants, respectively. The regression analysis found that the mean change in HbA1c from baseline to follow-up was not statistically significant across groups (p=.30); however, there was a significant improvement within the intervention group. In a *post hoc* analysis (paired t-test) the intervention group showed a significant decrease in HbA1c at follow-up (–0.95%, p<.05) when compared with baseline (Figure 2).

The proportion of patients with HbA1c levels of less than 7% three months after baseline was significantly improved in the intervention group compared with the control group. Patients with at goal levels of HbA1c (< 7%) during baseline were 29% and 32% in the control and intervention groups, respectively. However, at follow up 58% of patients assigned to the intervention group had HbA1c levels at therapeutic goal compared with only 21% in the control group (p <.05) (Figure 3).

*Blood pressure*. Mean systolic blood pressure (SBP) and diastolic blood pressure (DBP) was 130 ± 4 mmHg and 81 ± 3 mmHg, respectively, for all participants at baseline (n=43). Forty-six percent did not meet ADA target blood pressures of less than 130/80 mmHg and 28% had systolic blood pressure (SBP) ≥ 140 mmHg and/or diastolic blood pressure (DBP) ≥ 90 mmHg. There was no significant reduction in systolic or diastolic blood pressure for control and intervention group participants three months post-baseline (p>.05) (Table 2).

*Random blood glucose*. Mean random blood glucose (RBG) was 183 ±15 and 190 ±23 mg/dl for control and intervention group participants, respectively, at baseline (Table 2). Thirty-four percent of all participants had RBG greater than 200 mg/dl and 20% had RBG greater than 250 mg/dl at baseline. Seven-percent (n= 3/43) had

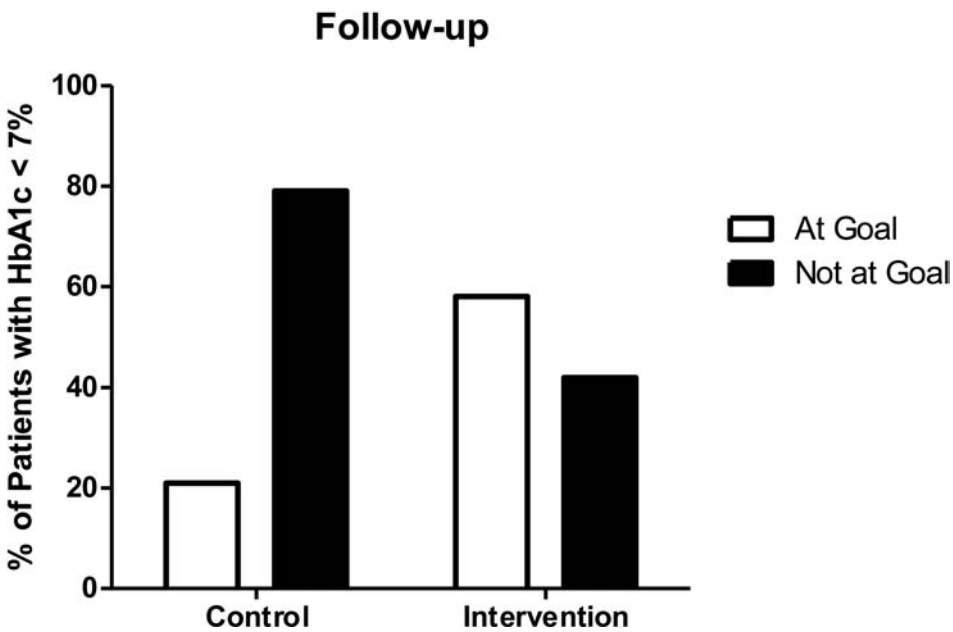


Figure 3. Percentage of patients in the control and intervention groups that showed hemoglobin A1c levels of less than 7% (at goal) or more than 7% (not at goal) at follow up.

RBG greater than 400mg/dl and were referred to the local hospital emergency room with screening results. There were no significant changes in mean RBG for control or intervention group participants at follow-up (p>.05) (Table 2).

*Lipid levels and kidney function*. Participants in both groups had similar lipid levels at baseline. No significant changes on these clinical outcomes were observed at follow-up between groups (p>.05 Table 2). Regarding kidney function, five participants (12%) out of the 40 patients that were tested for evidence of kidney disease had evidence of microalbuminuria or proteinuria (> 30 mg/g albumin/creatinine ratio) as determined by a dipstick test.

## Discussion

This pilot study demonstrates that a short pharmacist intervention on Latino diabetics can be successful at improving clinical outcomes such as HbA1c and weight loss. It is known that for every percentage point decrease in HbA1c (e.g., 9 to 8%), there is a 25% reduction in diabetes-related deaths, a 7% reduction in all-cause mortality, and an 18% reduction in combined fatal and nonfatal myocardial infarction.[21] Weight loss can also have a significant impact on diabetic patients. Results from prospective studies confirm that obesity is a significant risk factor for increased morbidity and mortality from diabetes and cardiovascular disease, as well as for a broad range of comorbid conditions.[22,23] Meta-analyses of published literature have estimated that a weight gain of 1 kg increases coronary heart disease risk by approximately 3% in men and 6% in

women.[24] Weight management may be the most important therapeutic task for most obese T2DM individuals. Therefore, the results of this pilot study are relevant for diabetes management in a low-income and medically underserved Latino population.

Community pharmacists are among the most accessible health care professionals and can affect the prevention and treatment of chronic conditions such as diabetes by providing patient education, medication management, and point-of-care testing.[8] Independently of the practice setting, pharmacists' interventions have shown a decrease in HbA1c levels by 1.0 ± 0.28 %.[25] Positive outcomes have been observed on pharmacist interventions that range from three months to a year follow-up period.[26] The average time of the pharmacist–patient interaction is 35 ± 17 minutes; frequency is monthly in 50% of the studies and ranges from every two weeks to every three months.[8] The pharmacist intervention in this study was original since it included a three-hour focus discussion group session, and two face-to-face one hour sessions during the first six weeks of the study. Furthermore, the intervention was comprehensive, including medication therapy management, diabetes education, and lifestyle changes such as nutrition and physical activity. This intensive approach may explain the significant weight loss between groups and the decrease in HbA1c in the intervention group at follow-up for such a short intervention. Additionally, the intervention allowed an increase in pharmacists' exposure as well as awareness about their expanded role among the Latino community.

Randomized controlled studies evaluating pharmacist interventions targeting weight loss are scarce.[27] However, some initiatives have demonstrated that pharmacists can successfully manage weight loss in the community and health care centers.[27–29] Different strategies have been used by pharmacists to target weight loss including meal replacement, reduced-calorie diet or over the counter Orlistat with an average weight loss in 12 months of 1.1 to 4.1 kg.[27] We believe that the use of the plate method together with a Latin American Food Model kit in this study was a successful strategy used by the pharmacists to educate patients on portion sizes and food selections.

The impact of pharmacist interventions at improving clinical outcomes in Latino diabetics can be very relevant since this population is underserved, lacks health care access, and consequently has a higher prevalence of patients not at target for this disease.[4] Moreover, ethnic and culturally competent interventions that consider family support, language barriers, socioeconomic status, and health care access may decrease self-perceived barriers to self-care.[13] Some key factors contributed to the positive clinical outcomes findings in this study: 1) the PARDS intervention was conducted in Spanish and by Latino pharmacists; 2) cultural health beliefs, socioeconomic status, and health care access were considered; 3) action planning on physical activity and nutrition was tailored to each patient in order to set realistic goals; 4) a support person/family member was included, when available; and 5) the study was conducted in a community setting in partnership with a community-based organization.

Several challenges were identified in this study sample, including their limited access to health care services, high unemployment rates, and challenges communicating with providers. Participants' lack of health insurance significantly affected the pharmacist intervention outcomes. Many subjects were in need of a prescription for their diabetes, blood pressure, and/or cholesterol medications and had not had continuous access to health services and periodic monitoring by a primary care provider. They often

reported relying on foreign medication supply and/or access to informal networks (family, friends) to obtain drug treatment, supplies, and advice.

Some of the limitations of our study are the size of the sample and length of pharmacist intervention. Even though the attrition rate was high, a larger sample size and a longer pharmacist intervention would have resulted in more significant clinical findings. Furthermore, since the study was conducted in South Florida and patients were recruited by a community organization serving underserved Latinos in Broward County, the results may not apply to other Latino populations in the U.S. and may not be a good sample of the Latino population in South Florida.

In conclusion, this pilot study demonstrated that in a Latino population with limited health care access and facing multiple personal and environmental barriers, a short, innovative, culturally-competent tailored pharmacist intervention significantly improved diabetes control and promoted weight loss. These findings support the role of the pharmacist as a vital health care provider among Latino patients.

## Acknowledgments

The DLASST research project was funded by the Centers for Medicare & Medicaid Services—CMS 1HOCMS030309-02. Our special thanks to Dr. Richard Bragg, CMS Project Officer, for his ongoing support and feedback throughout the project. We also would like to thank Luis Felipe Pinzón, Hispanic Unity of Florida (HUF) Vicepresident of Programs and Magaly Alvarado, HUF Program Manager, for their ongoing commitment and support at the community level. The authors are grateful to the research participants, pharmacy student assistants, the participating community organization HUF, and the support of Nova Southeastern University College of Pharmacy.

**Disclaimer:** The views expressed on this manuscript do not represent the views of HRSA of U.S. government.

## Notes

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